

Choir: an antenna array that trusts what decodes

MATLAB in Space Hackathon, Track 1: Deep-Space Communication & Signal Intelligence

8 simulated dishes receive one weak spacecraft signal carrying real NASA Curiosity (MSL) telemetry values

The receiver finds the carrier, lines the dishes up, decodes CCSDS frames, and recovers from hidden faults on its own

Key idea: learn how much to trust each dish only from frames that pass the checksum and carry our spacecraft ID

The problem

A deep-space signal is far too weak for one dish, so NASA's Deep Space Network combines several (arraying)
Lining them up blindly (JPL's SUMPLE: each dish vs. the sum of the others) follows whatever the dishes agree on
A louder signal that reaches every dish, like another spacecraft or a radio source in the beam, is then what they agree on
DSN demand exceeds supply by up to 40% at times (NASA OIG IG-23-016, 2023); the receiver must also run itself
Track 1 asks for: noisy radio in, automatic filtering and (re)acquisition, decoded telemetry out

How Choir works (all MATLAB)

Transmit: telemetry -> CCSDS-style frames (spacecraft ID, counter, CRC-16) -> ccsdsTMWaveformGenerator (BPSK, RRC, sync marker, randomizer)

Channel (hidden truth): 8 dishes with their own SNR, delay, phase drift; carrier drift; seeded faults

SEARCH: carrier found by squaring BPSK and summing spectra over dishes

ALIGN: sync-marker search over a full frame period; per-dish delays; frames from other spacecraft rejected by ID

DECODE: weights $w = Q^{-1} h$. h (where the spacecraft is) is learned only from verified frames;

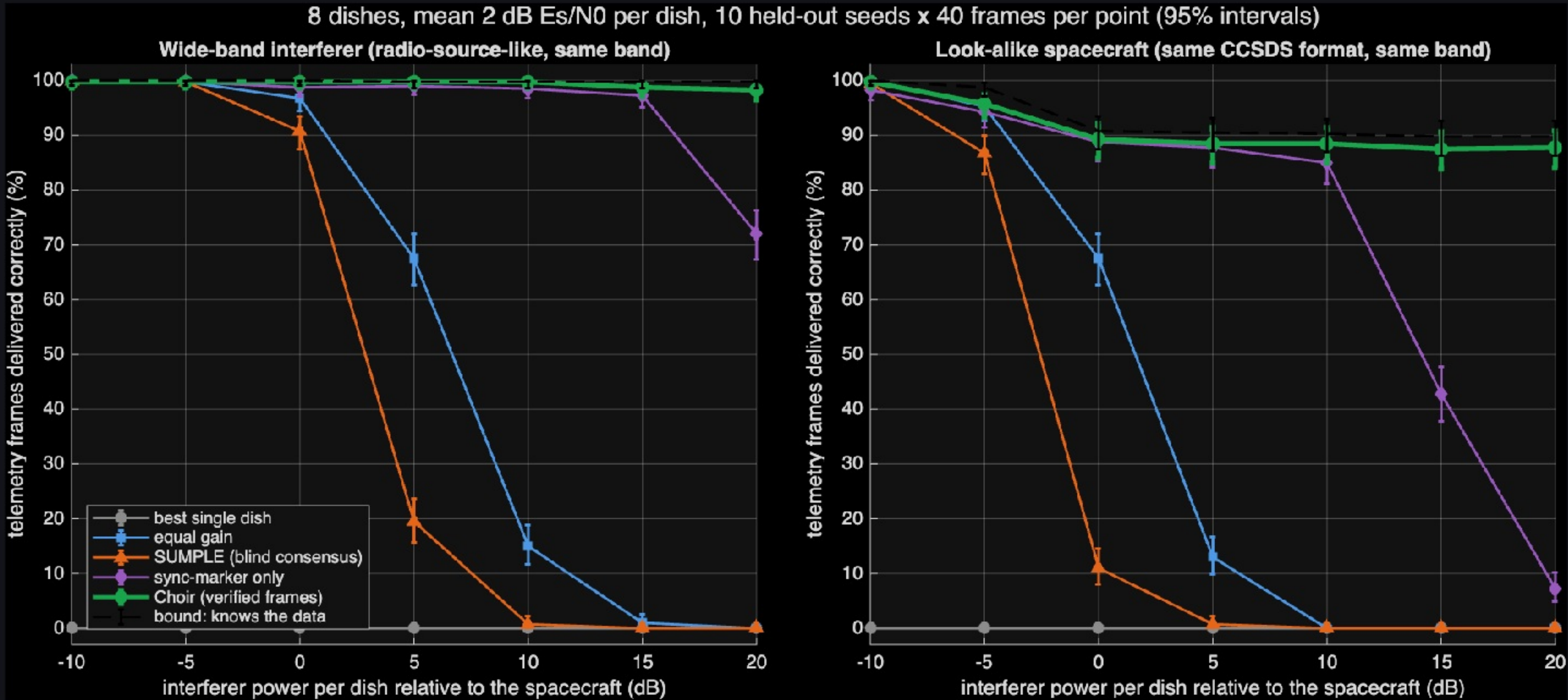
Q (everything else) comes from all samples, so new interference is cancelled without waiting for a good frame

Health: a dish that stops matching verified frames is re-timed or excluded, and probed until it returns

FALLBACK: 3 bad frames -> re-time every dish on the sync marker, equal weights; 8 bad -> full carrier re-search

Main result: interference that looks like the spacecraft

Look-alike spacecraft at +10 dB per dish: Choir 88%, SUMPLE 0%, equal gain 0%, sync-marker only 85% of frames correct
Wide-band source at +10 dB per dish: Choir 100%, SUMPLE 1%
False accepts across all interference runs: 0



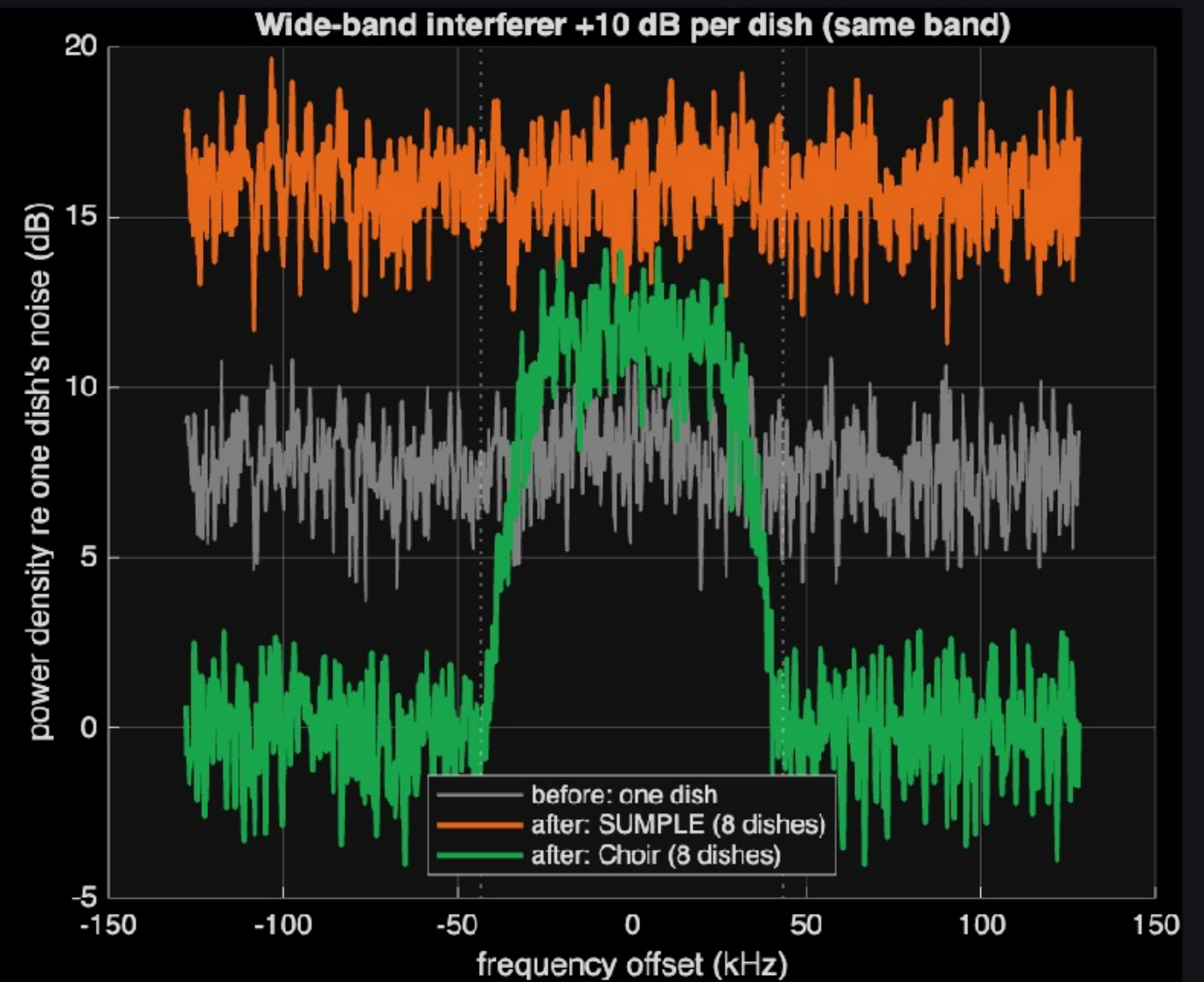
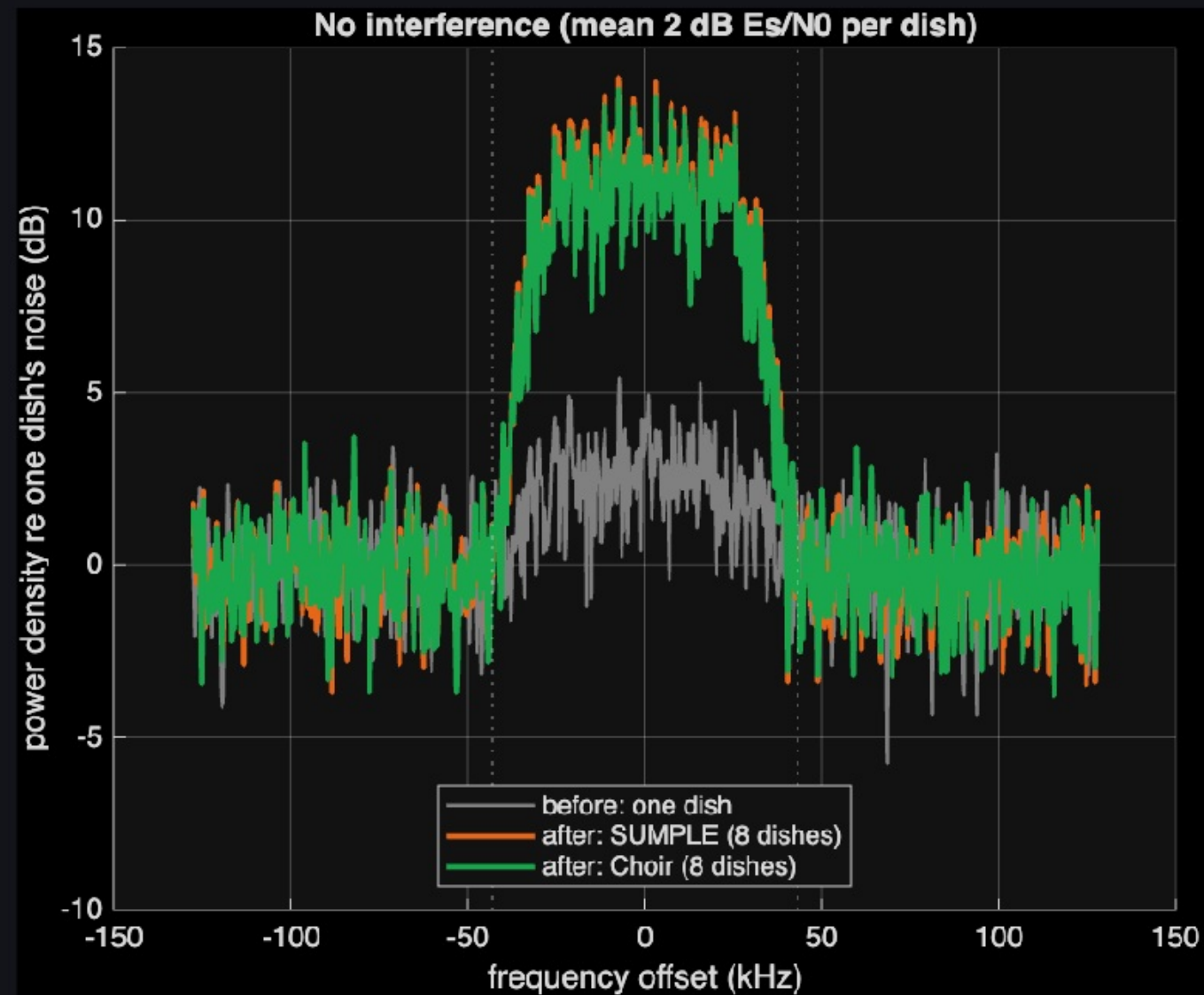
Autonomy: six hidden faults per run, nobody touches it

20 seeded runs, each with a dead dish, clock glitch, look-alike spacecraft, carrier hop, deep fade and noisy dish at random times
Correct frames: Choir 90.6% (worst run 80.7%); SUMPLE + same supervisor 77.7%; Choir with supervisor off 40.0%
Median recovery (Choir): carrier hop 284 ms, after deep fade 0 ms; false accepts: 0



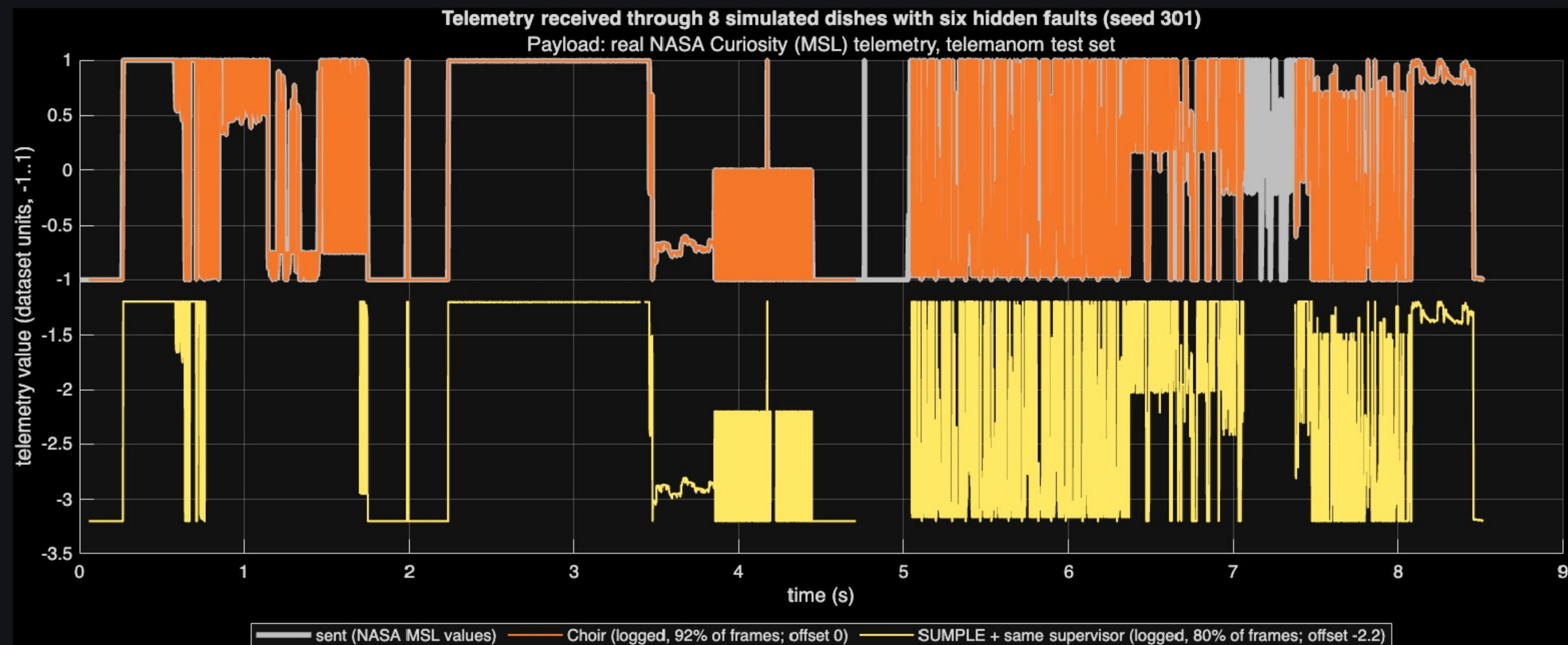
Before / after spectra (required Track 1 output)

Grey: one dish. Green: Choir's combined output. Orange: SUMPLE. Same noise scale.



Telemetry log: what actually arrived

Logged values from verified frames only; gaps are frames the receiver refused to log



Honest limits and next steps

The radio link is simulated; the telemetry values are real (MSL, telemanom dataset). No real multi-dish recording was available

Uncoded BPSK: real deep-space links use turbo/LDPC codes, which would shift every curve left by several dB

Interferers are modeled near the spacecraft's direction (shared per-dish delay); far off-axis ones need space-time weights

The checksum protects against noise, not a deliberate spoofer (that needs authenticated frames, e.g. CCSDS SDLS)

Choir needs some verified frames first: interference present before any frame ever verifies is its weak case

Next: elastic arraying (release dishes when margin allows), coding, real SatNOGS recordings, Stateflow supervisor